

MG 3 machine gun

The **Rheinmetall MG 3** is a German general-purpose machine gun chambered for the 7.62×51mm NATO cartridge. Manufactured by Rheinmetall for the *Bundeswehr*, designed and derived from the World War II era MG 42 that fired the 7.92×57mm Mauser round.^[2]

The MG 3 was standardized in the late 1950s and adopted into service with the newly formed *Bundeswehr*, where it continues to serve to this day as a squad support weapon and a vehicle-mounted machine gun.

The MG 3 and its derivatives have also been acquired by the armed forces of over 40 countries. Production rights to the machine gun were purchased by Italy (MG 42/59), Greece, Iran (as MGA3) Pakistan (as the MG 1A3), Sudan, Spain, and Turkey.^[3]

History

At the end of World War II, the original technical drawings and data for the 7.92×57mm Mauser-chambered MG 42 were captured by the Soviets. These would later be taken to Czechoslovakia and Yugoslavia. Rheinmetall had to reverse engineer the first postwar machine guns from an original MG 42 machine gun.^[4]

Production of the first postwar variant of the MG 42 chambered for 7.62×51mm NATO ammunition (designated the **MG 1**) was launched in 1958 at the Rheinmetall arms factory as requested by the Bundeswehr. Shortly thereafter, the machine gun was modified, receiving a chrome-lined barrel and

Rheinmetall MG 3



MG 3 7.62×51mm NATO general-purpose machine gun

Type	General-purpose machine gun
Place of origin	West Germany
Service history	
In service	1959–present
Used by	See <i>Users</i>
Wars	See <i>Conflicts</i>
Production history	
Designed	1959
Manufacturer	Rheinmetall License-built by: Beretta, MKEK, Ellinika Amyntika Systimata, Defense Industries Organization, Military Industry Corporation, Pakistan Ordnance Factories, General Dynamics Santa Bárbara Sistemas
Produced	1959–present
No. built	1 million+
Variants	See <i>Variants</i>
Specifications	
Mass	11.5 kg (25.35 lb) ^[1] 27.5 kg (61 lb) (mounted on tripod)
Length	1,225 mm (48.2 in) 1,097 mm (43.2 in) (without stock)

sights properly calibrated for the new round; this model would be named the **MG 1A1** (also known as the **MG 42/58**).

The **MG 1A2** (also known as the **MG 42/59**) is a further development of the MG 1A1, which has a heavier bolt (950 g (33.51 oz) for a slower 700–900 rounds per minute cyclic rate of fire, compared to 550 g (19.40 oz)), and a new friction ring buffer made suitable for using the heavier bolt. The MG 1A2 added new bolt-bounce preventing bolt catches to the action to resolve the ammunition ignition timing sensitivity of the preceding variants and was adapted to use both the standard German non-disintegrating *Patronengurt DM1* ammunition belt and the American M13 disintegrating belt. The **MG 1A3** features further improvements to the muzzle device, bipod and bolt.

Simultaneously, wartime 7.92×57mm Mauser chambered MG 42 machine guns that remained in service were converted to the standard 7.62×51mm NATO chambering and designated **MG 2**.

In 1968, the MG 3 was introduced and entered production. Compared to the MG 1A3, the MG 3 features an improved feeding mechanism with a belt retaining pawl to hold the belt up to the gun when the top cover plate is lifted, an added anti-aircraft sight and a new ammunition box. MG 3s were produced for Germany and for export customers by Rheinmetall until 1979. The preceding non-MG 3 variant machine guns in the *Bundeswehr* inventory were gradually converted to the MG 3 standard. Some additional production of the MG 3 in Germany was carried out by Heckler & Koch.^[5] The MG 3 and its variants all share a high level of parts interchangeability with the original MG 42.

MG 3s continue to be produced in Turkey and Pakistan.^[6] In 2019 there were plans in Germany to produce several thousand new MG 3 receivers to keep using vehicle mounted MG 3s in the low level anti-aircraft (designated MG 3A0A1) and turret mounted (designated MG 3A1A1) roles in the near future.^[7]

<u>Barrel length</u>	565 mm (22.2 in)
<u>Cartridge</u>	7.62×51mm NATO
<u>Action</u>	Recoil-operated, <u>roller locked</u>
<u>Rate of fire</u>	800–950 rounds/min (heavy-weight bolt) 1,000–1,200 rounds/min (standard-weight bolt)
<u>Muzzle velocity</u>	820 m/s (2,690 ft/s)
<u>Effective firing range</u>	200–1,200 m sight adjustments
<u>Maximum firing range</u>	600 m (1,969 ft) (bipod) 1,200–1,600 m (3,937–5,249 ft) (tripod mounted) 3,000 m (9,843 ft) (gun carriage) 3,750 m (12,303 ft) (terminal)
<u>Feed system</u>	50-round non-disintegrating DM1 belt (can be combined in a drum); 100-round disintegrating DM6/M13 belt
<u>Sights</u>	Open tangent <u>iron sights</u> or optical sights



Soldiers of the West German Bundeswehr on exercise in 1960. Pictured is the predecessor to the MG 3 —the MG 1A3 variant. The soldier on the right is carrying a G3 battle rifle. In the rear stands a SPz 11-2

Deployment

The MG 3 is still used as the standard secondary weapon of most modern German armoured fighting vehicle designs (e.g. Leopard 2, PzH 2000, Marder), as a primary weapon on light/non-armoured vehicles (e.g. LKW 2to, MAN gl-trucks, ATF Dingo) and as an infantry weapon on light bipods as well as different tripods. Since 2015, the German Armed Forces have supplemented the MG 3 with the Heckler & Koch MG5.

Reliability

In 1974, the US Army tested German made MG 3s alongside eight other contemporary GPMG designs to replace the then-in-service M219 Tank Machine Gun, which was considered unacceptably unreliable by the US Army. The MG 3 had worse results in mean rounds between failure (where it was the third worst entrant), mean rounds between stoppage (where it was the fourth worst entrant), cookoff rate (where it was the worst entrant), and accuracy (where it was the worst entrant) than the M60 and FN MAG. The only categories in which the MG 3 outperformed both the M60 and FN MAG were in ease of training, ease of changing barrels, and safety. The MG 3 was overall sixth of nine candidates in an unweighted ranking, and it was considered that no reasonable combination of attribute rankings existed in which it would be the winner of the contest.^[8]

Operational use

The German military instructs that sustained fire must be avoided at all costs. In the bipod mounted light machine gun role, MG 3 users are trained to fire short bursts of three to five rounds and strive to optimize their aim between bursts fired in succession. In the tripod mounted medium machine gun role, MG 3 users are trained to fire both short bursts and longer bursts of 20 to 30 rounds and strive to optimize their aim between successive bursts.^{[9][10]} The Bundeswehr trains soldiers to replace the barrel of the MG 3 after 150 live rounds (or 100 blank rounds) after sustained heavy fire, with

a new, cooler one; only once the barrel is hand-warm (able to be held with the bare hand for 30 seconds) can a barrel be reused. Replacing the barrel is a simple procedure on the MG 3. Non-observance of this technical limitation renders the barrel prematurely unusable. Care must be taken when replacing the barrel as after extended cyclical fire, the barrel can be dangerously hot, potentially approaching white hot. The machine gun crew member responsible for a hot barrel change is issued protective asbestos gloves or a cloth to prevent burns to the hands.^[10] The effective rate of fire is about 250 rounds per minute.^[11] Some other machineguns have thermally insulated grips attached to their quick-change barrel, at least going back to the ZB vz. 26 design of 1926.



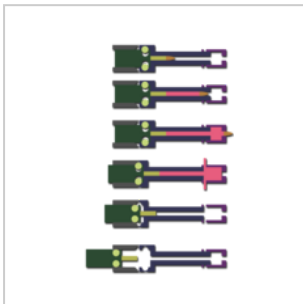
German soldier and U.S. Marine training with an MG 3

Design details

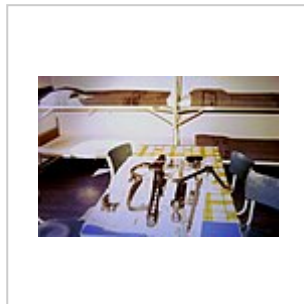
The MG 3 has an automatic-only trigger mechanism and a cross-bolt safety in the form of a button that is operated by the shooting hand (in its "safe" position the bolt release is disabled). The MG 3 fires from an open bolt. The cyclic rate can be altered by installing different bolts and recoil springs. A heavier bolt uses more recoil energy to overcome inertia, thus slowing the action. On MG 3 machine guns, two types of bolts are available, with standard weight (about 650 g (22.93 oz)) for the standard 1,000–1,200 rounds per minute cyclic rate of fire^[12] and with extra weight (about 900 g (31.75 oz)) for a slower 800–950 rounds per minute cyclic rate of fire. Those bolts also are used along with different return springs.^[4]

Operating mechanism

The MG 3 is an automatic, air-cooled, belt-fed short recoil–operated firearm. It features a roller locked bolt mechanism that consists of the bolt head, a pair of rollers, the striker sleeve, bolt body and return spring. The bolt is locked securely by a wedge-shaped striker sleeve, which forces two cylindrical rollers contained in the bolt head outward, and into corresponding recesses in the extension of the breech of the barrel. On firing, both the barrel and barrel extension recoil to the rear. The resulting impact (much like a Newton's cradle) moves the carrier to the rear, withdrawing the wedge and both rollers as they are cammed inward and out of their sockets by fixed cams, unlocking the bolt head. The bolt carrier and bolt then continue to the rear together guided by fixed guides while the barrel and barrel extension return to battery. Upon return of the bolt forward, the impact of the rollers against the camming surfaces on the breech carry the rollers from their seats, and, together with the surfaces on the striker sleeve, force the rollers outward, locking the bolt head into the barrel extension and ensuring a complete lock. The bolt also houses a spring-loaded casing extractor and ejector. Ejection is carried out when the ejector strikes the buffer head, sending a push forward through the ejector bar, which hits the ejector pin. This pin pushes the top of the base of the cartridge, which is still held by the extractor at the base, causing the empty casing to rotate and eject downward through the ejection chute.



MG 3 roller-locked boosted short recoil action diagram



Parts of a German MG 3



Markings on an original MG 42 retrofitted to a MG 3

Barrel

The MG 3 has a quick-change, chrome-lined barrel with four right-hand grooves and a rifling twist rate of 1 in 305 mm (1:12 in) and weighs 1.7 kg (3.7 lb). Alternatively, MG 3 barrels can also have polygonal rifling. The barrel is integrated with the barrel breech. During sustained firing, there is a need for the barrel to be changed and this is how they are swapped: The gun is cocked and the barrel catch on the right of the barrel shroud is swung forward. Then, the breech end of the hot barrel swings out and can be removed by elevating or twisting the gun. A fresh barrel would be inserted through the barrel catch and the muzzle bearing. When the catch is rotated back, the barrel is locked and the machine gun can resume

firing. Both the receiver housing and ventilated barrel casing are made from pressed sheet steel. The machine gun crew member responsible for a hot barrel change is issued protective asbestos gloves to prevent getting burned. A muzzle device is mounted at the end of the barrel and it acts as a flash suppressor, muzzle brake and recoil booster.

Feeding system

The MG 3 feeds from the left side through a feed block using a metal 50-round drum that uses a non-disintegrating metallic-link DM1 belts (*Patronengurt DM1*) ammunition belts (which have links that wrap around the cartridge case and are linked by a coiling wire on each side) or disintegrating-link M13 or DM6 belts.

In the light machine gun role, the MG 3 is deployed with a 100-round (or 120-round in case of disintegrating belts) belt fitted inside a synthetic ammunition drum developed by Heckler & Koch that is latched on to the left side of the receiver. The rear wall of the drum is transparent and serves as a visual indicator for the amount of ammunition available. The feed system operates through a feed arm that is housed in the feed cover. Two feed pawls are linked to the front end of the arm by an intermediate link and move in opposite directions, moving the belt in two stages as the bolt moves back and forward during firing.



MG 3 of the German Army

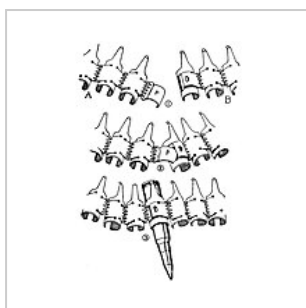
DM1 belts are intended for multiple reuse and in terms of design are based on and derived from the last version of the *Gurt 34/41*-belt family used in World War II in MG 34 and MG 42 machine guns.^[13] DM1 belts are preloaded at ammunition factories in 50-round connectable belt lengths and can be linked to any length necessary. Spent cartridge cases are ejected downwards, and the emptied links are transported to the right.

Alternatively, the MG 3 can also be fed by disintegrating metal M13 link belts (designated DM60 by Germany) used by many NATO member states. M13 links are also used on the Dillon M134D Minigun, M60, FN MAG, HK21 and MG5 machine guns among others. The disintegrating metal belt is fed from the left side. Ejection of empty M13 links is to the right side, and spent cartridge cases are ejected downwards. The inexpensive M13 links are considered disposable.

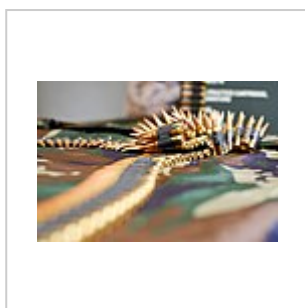
Both belt types are of the push-through type and use a metal lip that is arrested in the rim of the cartridges to correctly position and fix the cartridges in place. The feeding system is based on the direct push-through of the cartridge out of the belt link into the gun's chamber. Feed is performed in two steps by a pawl-type feeding mechanism that continues to move the belt during both the rearward and forward cycles of the reciprocating bolt, producing a smooth belt flow.

For field use, there are several ammunition containers available. The *Gurttrommel* (belt drum) contains a 50-round DM1 or DM60 belt. The *Gurttrommel* is not a true magazine but holds a curled 50-round belt preventing it from snagging, twisting and getting stuck during mobile assaults. The steel DM2 ammunition box contains a 250-round DM1 belt and the smaller plastic DM40004 ammunition box

contains a 100-round DM1 belt or a 120-round DM60/M13 belt. The German military tends to use non-disintegrating DM1 belts for general use and disintegrating DM60/M13 belts in vehicle or aircraft fixed MG 3 mountings that allow for collecting the ejected link pieces for reuse.



Method of joining German non-disintegrating metallic-link ammunition machine gun belts



Non-disintegrating metal DM1 belt



Disintegrating metal M13 link belt (designated DM60 by Germany)

Sights

The open-type iron sight line has a relatively short 430 millimeters (16.9 in) radius and consists of a "Λ-type" height adjustable front sight on a folding post and a leaf rear sight with an open V-notch sliding on a ramp, graduated from 200 to 1,200 meters (219 to 1,312 yd) in 100 meters (109 yd) increments. A flip-up anti-aircraft sight is attached to the receiver top just in front of the normal rear sight element.

Danish C79 LMG Optic

As an iron sighting line alternative, the Danish military uses a 3.4×28 optical sight mounted on top of the receiver which can be set from 300 to 800 meters (328 to 875 yd) in 100 meters (109 yd) increments on their MG 3 (designated as M/62) and later M/60E6. What sets the Danish C79 LMG Optic designated as M/98 apart from the standard C79 optical sight is its unique reticle of a chevron with a height of 12,5 TS and two TS line left and right with a length of 7,5 TS beginning 2,5 TS away from the tip of the chevron. The sight designated as M/99 is also available in a night vision configuration.^[14]

Bipod and tripod

For the light machine gun role, the MG 3 is equipped with a synthetic polymer stock and a folding bipod.

In the stationary, heavy machine gun role, the MG 3 is mounted on a buffered *Feldlafette* ("field tripod") that also features storage containers for accessories like the *Zielfernrohr 4 × 24* periscope-style telescopic sight. This direct fire sight is also mounted on the *Feldlafette* and is graduated from 0 to 1,600 meters (0 to 1,750 yd) in 100 meters (109 yd) increments. The *Zielfernrohr 4 × 24* reticle can be illuminated by an external unit. It can also be used with the FERO-Z 51 night sight.^{[9][10]}

A feature of the German World War II *Lafette 42* tripod that was not carried over to the MG 3 *Feldlafette* was the *Tiefenfeuerautomat* ("in-depth automatic fire"). If selected, this feature walked the fire in wave like motions up and down the range between predefined ranges. This sweeping of a given range (*Tiefenfeuer* – "in-depth fire") continued as long as the gun was fired.

Variants

- **MG 1:** Rheinmetall variant of the MG 42, most notably rechambered to fire 7.62×51mm NATO.
- **MG 1A1 (MG 42/58):** As MG 1, but with sights properly calibrated for the new round. Sights refitted to existing MG 1s.
- **MG 1A2 (MG 42/59):** MG 1A1 variant; product improved with longer ejection port, heavy bolt and friction ring buffer.
- **MG 1A3:** MG 1A2 variant; product improvement of all major components.
- **MG 1A4:** MG 1 variant; for fixed mount armour use.
- **MG 1A5:** MG 1A3 variant; MG1A3s converted to MG1A4 standard.
- **MG 2:** Designation for all wartime MG 42s rechambered to 7.62×51mm NATO.
- **MG 3:** MG 1A3 variant; product improved with AA rear sight.
- **MG 3E:** MG 3 variant; reduced weight model (roughly 1.3 kg lighter), entered into late 1970s NATO small arms trials.
- **MG 3A1:** MG 3 variant; for fixed mount armour use.
- **MG 3KWS:** MG 3 variant; developed by Rheinmetall and Tactics Group as a stand in until the HK121 replaces it.^[15]
- **MG 42/59:** Italian variant produced by Beretta, Whitehead Motofides and Franchi, since 1959, Chambered in 7.62×51mm NATO. The bolt weight was increased to 1,200 g (42.33 oz)) for a reduced 800 rounds per minute cyclic rate of fire. Used mainly mounted on vehicles and has largely been phased out by the FN Minimi.
- **Ksp m/94:** Swedish variant chambered with the 7.62×51mm NATO round. Mainly used as secondary armament in Stridsvagn 122.



MG 3 seen here in the stationary, heavy machine gun role, mounted on a stabilized *Feldlafette* tripod and fitted with an optical *Zielfernrohr 4 × 24* periscope sight



MG 3 in the heavy machine gun setup on a *Feldlafette* tripod with mounted optical sight



Vehicle-mounted MG 3 at the Bundeswehr Day in 2019

Derivatives

The Rheinmetall RMG 7.62 is a mounted machine gun based from the Rheinmetall MG 3 with three rotating barrels (to reduce barrel erosion and overheating) is under development as a vehicle weapon. Only one barrel is active at a time: after one barrel overheats, it is rotated out for a cool one.

The MG14z is a double barrel derivative of the Rheinmetall MG 3 with two MG 3 receivers paired together. The MG14z enhances the firepower of military units that still issue the MG 3 or other MG 42 derivatives. It has been developed by the Tactics Group GmbH company as "a low-cost alternative to Miniguns".^[16]

Users

-  Afghanistan: Used by the Mujahideen and Taliban.^[17]
-  Albania: Used by the Albanian Army.^[18]
-  Argentina: Used by the Argentine Army.^[19]
-  Austria: Uses the **MG 74** which is a MG-42/59 variant licensed from Beretta and manufactured by Steyr Mannlicher. The MG 74's cyclic rate of fire is 850 rounds per minute.^[20]
-  Azerbaijan: The Military of Azerbaijan acquired a small quantity of MKEK MG 3s from Turkey.
-  Bangladesh.^[21]
-  Brazil: MG 3 and MG 3A1 are used by the Brazilian Army at the Leopard 1A5 BR.^[22]
-  Canada: Only used on 20 Leopard 2A6M CAN tanks acquired from Germany.^[23] Leopard 2s acquired from other sources will continue to use the FN MAG.
-  Cape Verde.^[24]
-  Chile.^[25]
-  Cyprus.^[26]
-  Czech Republic: Used on Dingo 2.^[27]
-  Denmark: MG-42/59 designated **M/62** in Danish service.^[28]
-  Estonia.^[29] Designated as **MG 3**.
-  Finland: As the **7.62 KK MG 3**. Used with the Leopard 2 tanks and NH90 helicopters.^[30]
-  Germany: Used by the Bundeswehr.^[24]
-  Ghana
-  Greece: License production by Hellenic Defence Systems (also known as EAS).^{[4][19]}
-  Iceland: Used by the Icelandic Coast Guard.^[31]
-  Indonesia: License production.^[32]
-  Iran: License production by Defense Industries Organization as the **MGA3**. Used by Iranian Army. Installed on Zulficar MBT.^{[25][33]}
-  Kurdistan: Peshmerga of Iraqi Kurdistan autonomous region, 40 MG 3s supplied by Germany and 100 MG-42/59 by Italy.^[34]
-  Italy: License production of the MG-42/59 by Beretta with parts made by Whitehead Motofides and Luigi Franchi; while largely replaced in squad support weapon role by the Belgian FN Minimi, it still sees widespread mounted use on ground-based vehicles and helicopters.^{[25][35]} Prior to the procurement of the Minimi, the *Stabilimento Militare Armi Leggere* (SMAL) at Terni has developed a kit to adapt the Italian Army's existing MG-42/59 machine guns to accommodate 5.56×45mm NATO ammunition. The kit comprises a new barrel, bolt head, feed opening and cover, recoil-enhancing element and a lighter bolt. The



Map with MG 3 users in blue



Austrian Army soldiers with an MG 74 and a Steyr AUG during the 2014 Combined Resolve II training exercise



Italian soldier with an MG-42/59 produced by Beretta

weight of the modified 5.56 mm MG-42/59 machine gun remains unchanged from the original version.

-  Latvia: used by the Latvian Land Forces.
-  Libya^[32]
-  Lithuania: used by the Lithuanian Armed Forces.^[36]
-  Mexico: License produced by SEDENA in Mexico.^[37]
-  Myanmar^[24] Clones made in Myanmar as the MA 15.^[38]
-  Niger^[39]
-  Nigeria^[40]
-  Norway^[24] In use by the Home Guard and Navy. Replaced in Army service by the FN Minimi and FN MAG in 2019.
-  Pakistan: Used by the Pakistan Army. Manufactured under license by Pakistan Ordnance Factories from 1960s in Wah Cantt.^[41]
-  Philippines: Used by the Philippine National Police.^[42]
-  Poland: On Leopard 2 tanks and support vehicles. Intended to be replaced by UKM-2000 and WKM-B.^[43]
-  Portugal^[25]
-  Sao Tome and Principe^[24]
-  Saudi Arabia^[24]
-  Slovakia: Mounted on Leopard 2A4.^[44]
-  Spain: Made under license.^[25]
-  Sri Lanka Used by the Sri Lankan Armed Forces in limited numbers.
-  Sudan: Made by Military Industry Corporation as the Karar.^[45]
-  Sweden: As the **KSP m/94**. Used with the Leopard 2 tanks.^[46]
-  Thailand: Used on Commando V-150 APC.
-  Togo^[24]
-  Tunisia^[32]
-  Turkey: Made by MKEK in Kırıkkale under license since 1974^[3] for the Turkish Armed Forces and Turkish Gendarmerie.^[47]
-  Ukraine: Ukrainian armed forces have been given MG 3 to aid them in the Russian invasion of 2022.^[48] At least 130 MG 3 have been provided by Germany.^[49]
-  Uruguay^[50]



Italy employs the MG 42/59 primarily on vehicles and rotary-wing aircraft. Seen here mounted on the B1 Centauro wheeled tank destroyer



A pintle-mounted MG 3A1 on a Norwegian Leopard 1 armoured recovery vehicle



MG 3 manufactured under license by Pakistan Ordnance Factories



Spanish Marines with an MG 3 manufactured by General Dynamics Santa Bárbara Sistemas

- Yemen: pro-Saudi forces and Al-Qaeda in the Arabian Peninsula.^[51]



Ukrainian Naval Infantry with an MG 3 [↗] during the Russian invasion of Ukraine

Non state users

- ISIL^[52]
- Provisional IRA^[53]

Former users

- Australia: The MG 3 was used between 1976 and 2007 as an anti-aircraft weapon on the Australian Army's Leopard AS1 MBT.^[54]

Conflicts

- Myanmar Conflict (1948–present)
- Nigerian Civil War (1967–1970)
- The Troubles (Late 1960s–1998)
- Bangladesh Liberation War (1971)^[55]
- Lebanese Civil War (1975–1990)
- Kurdistan Workers Party Insurgency (1978 - present)
- Afghan Conflict (1978–present)
- Iran–Iraq War (1980–1988)
- Second Sudanese Civil War (1983–2005)^[56]
- Somali Civil War (1988–present)
- Afghan Civil War (1992–1996)
- Kargil War (1999)
- Iraqi Conflict (2003–present)^[57]
- Insurgency in Khyber Pakhtunkhwa (2004–present)
- Boko Haram insurgency (2009–present)^[39]
- Libyan Crisis (2011–present)
- Syrian Civil War (2011–2024)
- Russo-Ukrainian War (2014–present)^[58]
- Yemeni Civil War (2015–present)^[59]
- Myanmar Civil War (2021–present)

See also

- MG51—7.5×55mm Swiss general-purpose machine gun
- SIG 710-3—Swiss derivative of MG 42

References

1. Bundesministerium der Verteidigung (26 June 1979). "Dv 3-14 Das Maschinengewehr" (http://upload.wikimedia.org/wikipedia/commons/b/bd/ZDv_3-14_Das_Maschinengewehr.pdf) (PDF) (in German) – via Wikimedia Commons.
2. Woźniak, Ryszard (2001). *Encyklopedia najnowszej broni palnej—tom 3 M-P* (in Polish). Warsaw: Bellona. p. 106.
3. "7.62 mm MG3 Machine Gun" (<https://web.archive.org/web/20090915211807/http://www.mkek.gov.tr/english/foUrunDetaylari.aspx?iKodUrun=90&iKodUrunKategorisi=122>). MKEK. Archived from the original (<http://www.mkek.gov.tr/english/foUrunDetaylari.aspx?iKodUrun=90&iKodUrunKategorisi=122>) on 15 September 2009. Retrieved 4 April 2014.
4. Moss, Matthew. "Rheinmetall Machinengewehr 3 West Germany" (<https://www.historicalfirearms.info/post/156145896479/rheinmetall-machinengewehr-3-west-germany>). *Historical Firearms*.
5. Shepherd, Christian (17 January 2009). "Machine Gun 42" (<https://web.archive.org/web/20120707064206/http://www.tactical-life.com/online/tactical-weapons/machine-gun-42/>). *Tactical Life*. Archived from the original (<http://www.tactical-life.com/online/tactical-weapons/machine-gun-42/>) on 7 July 2012. Retrieved 9 February 2014.
6. Gao, Charlie (24 August 2019). "Bang: Meet the 5 Best Machine Guns on the Planet" (<https://nationalinterest.org/blog/buzz/bang-meet-5-best-machine-guns-planet-75816>). *The National Interest*.
7. "Harmonisierung bei den Handwaffen der Bundeswehr" (<https://esut.de/2019/03/fachbeitraege/ruestung/11002/harmonisierung-bei-den-handwaffen-der-bundeswehr/>). *Europäische Sicherheit & Technik* (in German). 6 March 2019.
8. Attribute Analysis of the Armor Machine Gun Candidates (<https://apps.dtic.mil/sti/pdfs/ADA018625.pdf>) (PDF) (Report). US Army Armaments Command. July 1975. Archived (<https://web.archive.org/web/20210914174830/https://apps.dtic.mil/sti/pdfs/ADA018625.pdf>) (PDF) from the original on 14 September 2021. Retrieved 14 September 2021.
9. A2-222/0-0-4750 *Schießen mit Handwaffen (Shooting with handguns)* (<https://web.archive.org/web/20220817101158/https://www.reservistenverband.de/wp-content/uploads/2018/05/Zentralrichtlinie-Schie%C3%9Fen-mit-Handwaffen.pdf>) (PDF) (in German). Bundeswehr - Ministry of Defence Germany. 2015. Archived from the original (<https://www.reservistenverband.de/wp-content/uploads/2018/05/Zentralrichtlinie-Schie%C3%9Fen-mit-Handwaffen.pdf>) (PDF) on 17 August 2022. Alt URL (https://ia804708.us.archive.org/27/items/zentralrichtlinie-schiessen-mit-handwaffen-a-2-222-0-0-4750/Zentralrichtlinie-Schie%C3%9Fen-mit-Handwaffen%20A2-222_0-0-4750.pdf)
10. *ZDv 3/14 Das Maschinengewehr* (https://ia800703.us.archive.org/19/items/gunmanual_German_Zdv314_Das_Machinengewehr/German%20Zdv314%20Das%20Machinengewehr.pdf) (PDF) (in German). Germany: Bundeswehr (Ministry of Defence). 2003. Archived from the original (https://archive.org/details/gunmanual_German_Zdv314_Das_Machinengewehr) on 22 December 2022.
11. "Rheinmetall MG-3 - General-purpose machine gun" (<https://web.archive.org/web/20221209035642/http://www.military-today.com/firearms/mg3.htm>). Archived from the original (<http://www.military-today.com/firearms/mg3.htm>) on 9 December 2022. Retrieved 28 November 2022.
12. *Jane's Infantry Weapons*. Jane's Information Group. 2002. p. 2169.
13. Legendre, Jean-François (March 2005). "Feeding the Tiger: Ammunition Belts for German MG 34 and MG 42 Machine Guns" (<https://web.archive.org/web/20160522211121/https://www.smallarmsreview.com/display.article.cfm?idarticles=1897>). *Small Arms Review*. Archived from the original (<https://www.smallarmsreview.com/display.article.cfm?idarticles=1897>) on 22 May 2016.
14. Hjemmeværnet. "HRN-111-007" (<https://www.yumpu.com/da/document/read/17634649/hrn-111-007-762-mm-maskingevaer-m-62-hjemmevaernet>). *Hæren*: H-11.

15. Johnson, Steve (6 January 2014). "Bundeswehr MG3KWS Upgrade Program" (<https://www.thefirearmblog.com/blog/2014/01/06/bundeswehr-mg3kws-upgrade-program/>). *The Firearm Blog*. Retrieved 19 January 2014.
16. "Tactics Group MG-14z" (<https://web.archive.org/web/20141103092914/http://www.all4shooters.com/en/specials/Trade-shows-2014/IWA-2014/enforcetac/Tactics-Group-MG14Z/>). *All4shooters.com*. 8 March 2014. Archived from the original (<https://www.all4shooters.com/en/specials/Trade-shows-2014/IWA-2014-new-products/enforcetac/Tactics-Group-MG14Z/>) on 3 November 2014. Retrieved 28 June 2014.
17. McNab, Chris (2012). *MG 34 and MG 42 Machine Guns*. Osprey Publishing. p. 72.
18. "The 45 Albanian Army Special forces face-to-face with the Taliban's in Kandahar, Afghanistan" (https://web.archive.org/web/20190215041157/https://armyrecognition.com/almex_2011_daily_news_albania_defence_exhibition/the_45_albanian_army_special_forces_face-to-face_with_the_taliban_s_in_kandahar_afghanistan.html). *Armyrecognition.com*. 29 October 2011. Archived from the original (http://www.armyrecognition.com/almex_2011_daily_news_albania_defence_exhibition/the_45_albanian_army_special_forces_face-to-face_with_the_taliban_s_in_kandahar_afghanistan.html) on 15 February 2019. Retrieved 10 March 2022.
19. McNab, Chris (2012). *MG 34 and MG 42 Machine Guns*. Osprey Publishing. p. 70.
20. "Waffen und Gerät – Maschinengewehr MG 74" (https://www.bmlv.gv.at/waffen/waf_mg74.shtml). *Österreichs Bundesheer* (in German). Retrieved 9 February 2014.
21. "SALW Guide Global distribution and visual identification Bangladesh Country report" (<http://salw-guide.bicc.de/pdf/countries/023/bangladesh.std.en.pdf>) (PDF). Bonn International Center for Conversion. Retrieved 3 September 2019.
22. "Recebimento das metralhadoras para as viaturas Leopard 1A5 BR" (<https://web.archive.org/web/20100601094804/http://www.exercito.gov.br/NE/2010/05/10664/10664.pdf>) (PDF). *Noticiário do Exército* (in Brazilian Portuguese). No. 10, 664. Brasília: Centro de Comunicação Social do Exército. 21 May 2010. Archived from the original (<http://www.exercito.gov.br/NE/2010/05/10664/10664.pdf>) (PDF) on 1 June 2010.
23. Duchesne, Simon (14 February 2008). "Soldiers from C Squadron, Lord Strathcona's Horse (LdSH), wait for start time on their Leopard 2 tank" (http://www.combatcamera.forces.gc.ca/netpub/server.np?find&catalog=photos&template=detail_eng.np&field=itemid&op=matches&value=9381&site=combatcamera). *Canadian Forces Combat Camera*.
24. Jones, Richard D.; Ness, Leland S., eds. (27 January 2009). *Jane's Infantry Weapons 2009/2010* (35th ed.). Coulsdon: Jane's Information Group. ISBN 978-0-7106-2869-5.
25. "G3 Defence Magazine" (<http://en.calameo.com/read/000127853fed679f5ecec>). *Calameo.com*. 4 August 2010. Retrieved 9 February 2014.
26. "Ιερείς σε στρατιωτική άσκηση: Έπριξαν βολές και προκάλεσαν αντιδράσεις (BINTEO)" (<http://www.alphanews.live/cyprus/iereis-se-stratitiki-askisi-erixan-boles-kai-prokalesan-antidra-seis-binteo>). *Alphanews.live* (in Greek). 9 February 2022. Retrieved 15 January 2023.
27. "Obrněné vozidlo Dingo 2 CZ" (<https://web.archive.org/web/20140321223245/http://www.acr.army.cz/technika-a-vyzbroj/tanky/-obrne-vozidlo-dingo-2-cz-89947/>). *Armády České republiky*. Archived from the original (<http://www.acr.army.cz/technika-a-vyzbroj/tanky/-obrne-vozidlo-dingo-2-cz-89947/>) on 21 March 2014.
28. "Maskingeværer" (<https://web.archive.org/web/20170515135657/http://www2.forsvaret.dk/viden-om/materiel/udrustning/vaaben/Pages/Maskingev%C3%A6rer.aspx>). *Forsvaret*. Archived from the original (<http://www2.forsvaret.dk/viden-om/materiel/udrustning/vaaben/Pages/Maskingev%C3%A6rer.aspx>) on 15 May 2017. Retrieved 30 July 2015.
29. "Relvad – Kaitsevägi" (<https://mil.ee/kaitsevagi/tehnika/relvad/#t-762-mm-kuulipilduja-mg-3>). *Mil.ee*. Retrieved 14 July 2022.

30. "Puolustusvoimat" (<https://web.archive.org/web/20080423014838/http://www.mil.fi/maavoimat/kalustoesittely/index.dsp?level=63&equipment=217>). *Mil.fi* (in Finnish). Archived from the original (<http://www.mil.fi/maavoimat/kalustoesittely/index.dsp?level=63&equipment=217>) on 23 April 2008. Retrieved 9 February 2014.
31. "Hátt í 400 hríðskotabyssur í landinu" (<https://ruv.is/frett/hatt-i-400-hridskotabyssur-i-landinu>). *RÚV*. 27 October 2014. Retrieved 27 October 2014.
32. Jenzen-Jones, N.R.; McCollum, Ian (April 2017). *Web Trafficking: Analysing the Online Trade of Small Arms and Light Weapons in Libya* (<https://web.archive.org/web/20170516085143/http://www.smallarmssurvey.org/fileadmin/docs/F-Working-papers/SAS-SANA-WP26-Libya-web-trafficking.pdf>) (PDF). Working Paper No. 26. Small Arms Survey. p. 85. Archived from the original (<http://www.smallarmssurvey.org/fileadmin/docs/F-Working-papers/SAS-SANA-WP26-Libya-web-trafficking.pdf>) (PDF) on 16 May 2017.
33. "AIG: 7.62 mm Machin Gun" (https://web.archive.org/web/20081030031846/http://www.diomil.ir/en/aig.aspx?search_id=mga3). *Diomil.ir*. Archived from the original (http://www.diomil.ir/en/aig.aspx?search_id=mga3) on 30 October 2008.
34. Small Arms Survey (2015). "Trade Update: After the 'Arab Spring' " (<https://web.archive.org/web/20150701142546/http://www.smallarmssurvey.org/fileadmin/docs/A-Yearbook/2015/eng/Small-Arms-Survey-2015-Highlights-EN.pdf>) (PDF). *Small Arms Survey 2015: weapons and the world* (<http://www.smallarmssurvey.org/fileadmin/docs/A-Yearbook/2015/eng/Small-Arms-Survey-2015-Highlights-EN.pdf>) (PDF). Cambridge University Press. p. 110. Archived from the original (<http://www.smallarmssurvey.org/fileadmin/docs/A-Yearbook/2015/eng/Small-Arms-Survey-2015-Chapter-04-EN.pdf>) (PDF) on 1 July 2015.
35. "Armi Leggere" (https://web.archive.org/web/20110520181849/http://www.esercito.difesa.it/root/equipaggiamenti/armi_legg.asp#individuali). *Esercito Italiano: Il Portale delle Armi dei Materiali e dei Mezzi*. Archived from the original (http://www.esercito.difesa.it/root/equipaggiamenti/armi_legg.asp#individuali) on 20 May 2011.
36. "Kulkosvaidis MG-3" (https://web.archive.org/web/20151102060234/http://kariuomene.kam.lt/lt/ginkluote_ir_karine_technika/kulkosvaidziai/kulkosvaidis_mg-3.html). *Lietuvos kariuomenė* (in Lithuanian). 17 April 2009. Archived from the original (http://kariuomene.kam.lt/lt/ginkluote_ir_karine_technika/kulkosvaidziai/kulkosvaidis_mg-3.html) on 2 November 2015. Retrieved 18 October 2010.
37. "Mexican Army military equipment" (https://web.archive.org/web/20150614234009/http://www.armyrecognition.com/mexico_mexican_army_land_ground_forces_uk/mexico_mexican_army_land_ground_forces_military_equipment_armoured_vehicle_pictures_information_desc.html). *Army Recognition*. Archived from the original (http://www.armyrecognition.com/mexico_mexican_army_land_ground_forces_uk/mexico_mexican_army_land_ground_forces_military_equipment_armoured_vehicle_pictures_information_desc.html) on 14 June 2015. Retrieved 11 January 2014.
38. "活跃在缅甸内战中的德国MG3机枪" (<https://web.archive.org/web/20180803104154/http://www.twoeggz.com/news/6578454.html>) [German MG3 machine guns active in the Burmese Civil War]. *Twoeggz.com* (in Chinese). 22 January 2018. Archived from the original (<http://www.twoeggz.com/news/6578454.html>) on 3 August 2018.
39. de Tessières, Savannah (January 2018). At the Crossroads of Sahelian Conflicts: Insecurity, Terrorism, and Arms Trafficking in Niger (<https://web.archive.org/web/20180612184751/http://www.smallarmssurvey.org/fileadmin/docs/U-Reports/SAS-SANA-Report-Niger.pdf>) (PDF) (Report). Small Arms Survey. p. 58. ISBN 978-2-940548-48-4. Archived from the original (<http://www.smallarmssurvey.org/fileadmin/docs/U-Reports/SAS-SANA-Report-Niger.pdf>) (PDF) on 12 June 2018.
40. Jowett, Philip (2016). *Modern African Wars (5): The Nigerian-Biafran War 1967–70*. Oxford: Osprey Publishing. p. 20. ISBN 978-1-472816092.

41. "Machine Gun MG3 (MG1A3)" (https://web.archive.org/web/20131029205451/http://www.pof.gov.pk/IW_MG3.aspx). *Pakistan Ordnance Factories*. Archived from the original (http://www.pof.gov.pk/IW_MG3.aspx) on 29 October 2013. Retrieved 9 February 2014.
42. "PNP boosts capability after acquiring P1.871-B new equipment" (<https://mb.com.ph/2019/05/06/pnp-boosts-capability-after-acquiring-p1-871-b-new-equipment/>). *Manila Bulletin*. 6 May 2019.
43. Gaszak, Agnieszka (21 February 2011). "Pierwsze strzelanie żołnierzy NSR z karabinu maszynowego MG-3" (<http://www.wojsko-polskie.pl/narodowe-sily-rezerwowe/narodowe-sily-rezerwowe-aktualnosci/16521.pierwsze-strzelanie-zolnierzy-nsr-z-karabinu-maszynowego-mg-3.html>). *Wojsko Polskie*. Retrieved 10 May 2021.
44. *Ostré strelby z tankov Leopard 2A4 | Pozemné sily | Ozbrojené sily SR* (<https://www.youtube.com/watch?v=jEmlpssyWvI>), 14 August 2023, retrieved 31 August 2023
45. "KARAR 7.62x51mm" (<https://web.archive.org/web/20080310194442/http://mic.sd/images/products/wepons/en/KARARrsh.html>). *Military Industry Corporation*. Archived from the original (<http://mic.sd/images/products/wepons/en/KARARrsh.html>) on 10 March 2008.
46. Svensk, Henrik. "Strv122 Stridsvagn 122 / Leopard 2" (<http://www.soldf.com/strv122.html>). *SoldF.com*. Retrieved 9 February 2014.
47. "Silahlar" (<https://web.archive.org/web/20090414160114/http://www.jandarma.tsk.mil.tr/silahlar/silahlaric.htm>). *Gendarmerie General Command*. Archived from the original (<http://www.jandarma.tsk.mil.tr/silahlar/silahlaric.htm>) on 14 April 2009.
48. Nikolov, Boyko (28 March 2022). "1,500 Soviet Strela-2 MANPADS and 100 MG3 arrived in Ukraine" (<https://bulgarianmilitary.com/2022/03/28/1500-of-soviet-9k32-strela-2-manpads-and-100-mg3-arrived-in-ukraine/>). *BulgarianMilitary.com*.
49. "Militärische Unterstützungsleistungen für die Ukraine" (<https://www.bundesregierung.de/breg-de/themen/krieg-in-der-ukraine/lieferungen-ukraine-2054514>). *Presse- und Informationsamt der Bundesregierung* (in German). Retrieved 9 November 2022.
50. Porfilio, Gabriel (25 September 2020). "El Ejército de Uruguay recibe un millar de ametralladoras MG-3 adquiridas a España" (<https://www.infodefensa.com/texto-diario/mostr ar/3125084/ejercito-uruguay-recibe-millar-ametralladoras-mg-3-adquiridas-espana>). *InfoDefensa* (in Spanish). Retrieved 20 April 2022.
51. Austin, Rod (28 November 2018). "Yemen: inquiry finds Saudis diverting arms to factions loyal to their cause" (<https://www.theguardian.com/global-development/2018/nov/28/arms-yemen-militia-were-supplied-by-west-find-analysts>). *The Guardian*.
52. VICE News (27 April 2016). "What It's Really Like to Fight for the Islamic State" (<https://www.youtube.com/watch?v=aM3EITvF52I&t=1m52s>). *YouTube*. Archived (<https://ghostarchive.org/varchive/youtube/20211212/aM3EITvF52I>) from the original on 12 December 2021.
53. "Arms trial hears of 'dreaded weapon' ". *Irish Independent*. 5 February 1994.
54. Handel, Paul D. (2001). "Leopard AS1 Main Battle Tank Part One" (http://anzacsteel.hobbyvista.com/Armoured%20Vehicles/leopardph_1.htm). *Anzac Steel*. Retrieved 11 January 2014.
55. McNab, Chris (2002). *20th Century Military Uniforms* (2nd ed.). Kent: Grange Books. p. 31. ISBN 1-84013-476-3.
56. "Sudan – Global trade, local impact: Arms Transfers to all Sides in the Civil War in Sudan" (<https://www.hrw.org/sites/default/files/reports/sudan0898%20Report.pdf>) (PDF). *Human Rights Watch Report*. **10** (4): 25. August 1998.
57. VICE News (27 April 2016). "What It's Really Like to Fight for the Islamic State" (<https://www.youtube.com/watch?v=aM3EITvF52I&t=1m52s>). *YouTube*. Archived (<https://ghostarchive.org/varchive/youtube/20211212/aM3EITvF52I>) from the original on 12 December 2021.
58. Nikolov, Boyko (28 March 2022). "1,500 Soviet Strela-2 MANPADS and 100 MG3 arrived in Ukraine" (<https://bulgarianmilitary.com/2022/03/28/1500-of-soviet-9k32-strela-2-manpads-and-100-mg3-arrived-in-ukraine/>). *BulgarianMilitary.com*.

59. *Dnevnik* (<https://4d.rtvsl.si/arhiv/dnevnik/174353873>) (in Slovenian). Ljubljana, Slovenia: Radiotelevizija Slovenija. 2015. Event occurs at 17:13. Retrieved 8 August 2015.

Citations

- Ezell, Edward C. (1988). *Small Arms Today 2nd Edition*. Harrisburg, PA: Stackpole Books.
- Woźniak, Ryszard (2001). *Encyklopedia najnowszej broni palnej—tom 3 M-P* (in Polish). Warsaw: Bellona. ISBN 83-11-09311-3.

External links

- Bimbel.de—MG 3 disassembled (<http://www.bimbel.de/artikel/artikel-14.html>)
 - Bimbel.de—MG 3 on mount (<http://www.bimbel.de/artikel/artikel-26.html>)
 - MG 3 video (<https://www.youtube.com/watch?v=TAxkvctd5Ps>)
 - Video simulation of how a MG 3 works (https://www.youtube.com/watch?v=TfL_jlGyKCE) on YouTube
 - Machine Gun MG 3 (MG1A3) (https://web.archive.org/web/20131029205451/http://www.pof.gov.pk/IW_MG3.aspx)
 - ZDv 3–14 Das Maschinengewehr
 - Das Maschinengewehr MG 3 (<https://www.bundeswehr.de/de/ausruestung-technik-bundeswehr/ausruestung-bewaffnung/mg3>)
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